

Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06) held in
Uttarakhand, India

COMPARATIVE ANALYSIS OF LARGE LANGUAGE MODELS FOR THE APPLICATION OF SCIENTIFIC ARTICLE SUMMARIZATION

Rishabh Saxena^a, Shubhangi Singh^b, Preeti Dubey^c *

^{a,b,c}Dept. of Computer Science & Engg., SSET, Sharda University, India

Abstract - Automatic text summarization has become an essential tool for academics to stay up to date with the newest advancements due to the high rise of scientific publications. Abstractive text summarisation tasks can be done using LLMs. However current summarization methods often struggle to capture the nuanced and technical details present in research papers and there are prevalent research gaps in the realm of abstractive summarisation using generative AI. This study aims to analyse the efficiency of LLM models – GPT-3.5, LLaMa 3, Mixtral 8x7b and Gemma-2, in condensing detailed scientific literature into manageable summaries, and their further evaluation based on ROUGE, BERT and BLEU scores. The results show that GPT-3.5 and LLaMa 3 generate more coherent and contextually accurate summaries with a BERT score of 0.21 and 0.19 respectively, even though Mixtral 8x7B performs exceptionally well in quantitative measurements. Despite its coherence, Gemma-2 receives poorer performance in both qualitative and quantitative assessments. These findings bring out the value of integrating quantitative and qualitative evaluations to gain a thorough understanding of summarization.

© 2025 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06)

Keywords: LLMs; LLaMa-3; GPT-3; Mixtral; Gemma; ROUGE; BLEU; BERT; Automatic text summarization;

1. Introduction

Research papers are one of the most essential components of an academic study. For scholars trying to keep up with the most recent development, the increasing amount of scientific literature poses a significant difficulty. Automatic text summarization has become a crucial tool for tackling this problem, offering brief and logical summaries that capture the key points of lengthy research articles. This is specifically essential and useful for researchers who must effectively condense enormous volumes of data into understandable insights.

The success of automatic summarization has shown great promise in recent advances in Natural Language Processing (NLP), particularly the creation of Large Language Models (LLMs). The power of generative AI has significantly advanced the field of text summarization, enabling the creation of both extractive and abstractive summaries that were previously unattainable with traditional methods. LLMs with the use of generative AI improve the summarization process by comprehending and extracting complicated scientific texts thanks to their thorough training on a variety of large datasets. While both extractive and abstractive summarization approaches are effective, lesser work on abstractive summarization using generative AI has been done.

There are of course questions raised when it comes usage of AI tools over traditional research methodologies.

* Corresponding author. Tel.: +91-87870-34184 E-mail address: rishabh.sa676@gmail.com

Methods like summarization always bring with themselves some degree of information loss. A study done by Veen et al. [1] showed that 10 physicians assessed the summaries for their completeness, accuracy, and clarity. In most cases, the summaries generated by the LLMs were rated as either on par with or better than those written manually. Through the understanding of such studies and surveys, Langstron and Ashford [2] presented a system for the summarization multiple abstracts that showed strong perform in technical research such as healthcare and science. As more and more novel ideas have come to surface, they have been compiled into a single survey document for a better analysis of which application works best with LLM. One such survey was done by Sengar et al. [3] which took into account the various studies and industries LLM can me applied on such as medical diagnostics, text-image fusion, natural language processing as well discussing in comprehensive manner the latest advancement in all these fields.

Summarizing research papers is challenging due to their complex structure and dense content, which generic techniques often fail to handle. Generative AI, though promising, struggles with technical nuances and has limited application in academic literature. Summarizing tables, graphs, and data representations requires advanced interpretation and information extraction capabilities. Capturing a paper's contributions and findings demands combining abstractive and extractive methods, significantly increasing complexity. These challenges make summarization resource-intensive and highlight the need for more advanced solutions.

This paper analyzes the abstractive summarization abilities of LLMs—GPT-3.5, LLaMa 3, Mixtral 8x7B, and Gemma 2, for research paper summarization. Their performance is evaluated with metrics like BLEU [4], ROUGE-1, 2 & L [5], and BERT Score [6], which assesses semantic similarity, using the arXiv dataset [7]. The study compares the strengths and limitations of these models, identifying challenges they face with academic texts. Additionally, it explores potential applications and improvements to enhance their effectiveness for specialized tasks in academic and professional domains.

This paper begins with an introduction to text summarization challenges, followed by a literature review of prior work. The methodology details the generative AI models and evaluation metrics used. Results analyze model performance and insights, concluding with a summary and future research directions.

2. Literature Review

This section offers a thorough analysis of the major developments in summarization techniques, following the evolution from early efforts to SOTA approaches. We hope to clarify the current research and show the development of techniques by examining the groundbreaking work and more recent developments in this field.

Table 1: Literature Survey

Year	Description	Remarks
2024	Keswani et al. [8] present a novel approach that improves summarization and question answering by preventing LLMs from being overloaded with redundant information.	The methodology enhances the generation of effective summaries and answers, thereby improving the overall performance of LLMs.
2024	Yadav et al. [9] present their novel approach - TGETS, where nodes represent groups of sentences, and edges represent relation between sentences. The summaries are generated based on sentence weight and the average weight of the textual graph.	TGETS outperforms the existing PageRank method, achieving better precision, recall, and F1-scores for both R1 and R2 metrics in summaries of a couple 100 words.
2024	Aswani et al. [10] Explains the various corpora and evaluation techniques, examines transformer-based ways for summarizing scientific research articles, and highlights the field's limitations and future possibilities.	The research highlights the complexity of summarizing scientific texts due to their unique structure and reviews existing approaches, corpora, and evaluation methods.
2024	Jin et al. [11] review recent LLM-based ATS initiatives, providing an updated survey and practical insights from a point-of-sale perspective.	As the first survey to specifically investigate LLM-based ATS approaches, it offers valuable insights for both academic and industrial applications.
2024	Alli-balogun [12] describe a model that generates realistic images from	Using NLP to extract keywords for summarization and

	summarized text using Generative Adversarial Networks (GANs).	then utilizing BART transformers. The summarized text is then fed into the GAN network.
2024	Han et al. [13] propose TopicSum, an extractive summarization system using graph neural networks to leverage topic information and dynamically minimize redundancy.	The system outperforms forementioned extractive and abstractive algorithms in terms of Rouge scores when tested on the Pubmed, arXiv, and GovReport datasets.
2024	A study done by Liu et al. [14] proposes an improved method that combines the TextRank algorithm with K-Means clustering. Combining the TR scores, BM25 similarity, and the updated BM25 model enhances summary accuracy and cluster-center selection.	This approach outperforms other algorithms, including Lead-3, TextRank, and MBM25EMB, in ROUGE-1, ROUGE-2, and ROUGE-L metrics, improving accuracy and reducing redundancy in automatic text summarization.
2024	Sun and Platos [15] propose an enhanced encoder-decoder model with hierarchical attention and multi-objective reinforcement learning, addressing out-of-vocabulary issues via a pointer-generator network and improving semantic capture using multi-head self-attention.	Significant improvements in ROUGE metrics and semantic similarity of generated summaries to reference summaries.
2024	Yadav and Gopinathan [16] explore using Knowledge Graph Embeddings (KG Embed) for information retrieval and query extraction in scientific research across four repositories from 2000–2022.	One possible way to rank the articles that are closest to the term is to utilize KG Embed for query extraction and information retrieval in scientific research.
2024	Surjandari et al. [17] presents an improved email marketing by leveraging non-transactional data to personalize communications. It recommends modeling user preferences and segmenting clients according to subjects and prior email conversations. The Biterm Topic Model (BTM), which is used for topic modeling, performs well for short texts like email headers.	Using the BTM for topic modeling, as it is particularly effective for short texts. Research articles are longer, making it quite suitable for section of an article as a short text for topic modeling.
2024	Yadav et al. [18] propose a method for improving sentiment analysis by classifying phrases into positive or negative groups and using majority sentiment. They address language nuances like negation with the FFBC algorithm for better accuracy.	Applying FFBC to summary tasks allows for both the achievement of condensed content and the reflection of the authors' underlying sentiments or points of view.
2024	Glickman et al. [19] highlight the transformative impact of AI tools like ChatGPT, which enhance productivity by generating code, analyzing data, and modeling. LLM-based chatbots also simulate abductive reasoning to connect complex technical issues.	LLM-chatbots and AI tools streamline literature searches and research summarization, offering significant potential for enhancing productivity and innovation in statistics and data science.
2023	Pesaru et al. [20] developed a PDF chatbot using LangChain and an LLM model, enabling translation, text generation, and question answering from PDF data. It integrates Google Search for detailed responses and uses Pinecone for efficient document retrieval.	Demonstration of the potential of LangChain and LLM Models for creating informative chatbots across various domains, including customer service, education, and research.
2023	Nova [21] explores generative AI's healthcare applications, including summarizing EHR conversations, simplifying medical terms, and offering personalized care recommendations using data from wearables.	Personalized care recommendations leverage wearable data analysis with generative AI, offering tailored guidance on exercise, diet, sleep, and medication to enhance individualized care.
2023	Presented by Bandi et al. [22], this study aims to provide comprehensive insights for academics, developers, and practitioners in the field with an emphasis on requirements, models, input-output formats, and assessment metrics.	The study classifies input-output formats and discusses evaluation metrics to facilitate effective implementation, customization, and standardized assessment of generative AI systems.
2023	Heurta-Solo et al. [23] explore how machine learning can improve supply chain management (SCM), focusing on data related to transportation and financial transactions.	The study analyzes the impact of machine learning on sustainable supply chain management using MRAA to assess the correlation between keywords and outcomes.
2022	Kim et al. [24] propose a deep learning-based method for multi-patent summarization, using topic modeling, feature extraction, and a GAN summarization model to address the challenges of summarizing technical patent documents.	Tested on drone technology patents, the model outperformed existing DL models, offering high-quality summaries useful for Research and Development.
2022	Altmami et al. [25] review the state of the art in automatic summarization of scientific articles, highlighting recent approaches, corpora, evaluation	This study reviews the last decade's approaches, emphasizing extractive techniques and challenges from

	methods, limitations, and open problems.	the lack of benchmark corpora and gold standard summaries.
2021	Tarannum et al. [26] review popular approaches, pre-processing strategies, and trends in text summarization, highlighting that abstractive and hybrid methods yield the best results.	Abstractive and hybrid approaches lead to superior results in text summarization
2020	Gao et al. [27] proposes the Active Preference-based Reinforcement Learning (APRIL) framework and investigates preference-based interactive summarization in order to reduce the number of interaction rounds.	The system employs neural reinforcement learning for optimal summaries, preference learning for ranking, and active learning for user input.

The studies presented in Table 1 reflect the major advancements done in text summarization, knowledge graphs, and generative AI. While significant improvements have been achieved in metrics and domain-specific applications, challenges such as dataset scarcity and redundancy reduction persist, offering avenues for future innovation. The integration of summarization techniques with AI-driven tools showcases their expanding relevance across diverse disciplines.

3. Methodology

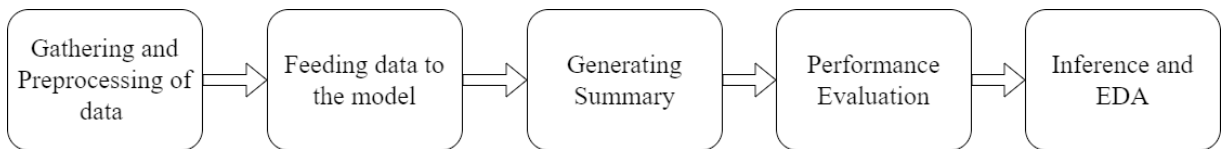


Fig 1. Methodology

After gathering and preprocessing the dataset, all four of the models were given the dataset as prompts for summarization. The summaries were then stored and used for evaluating the performance of the models using the mentioned evaluation metrics.

3.1. Data Preparation

We used a small sample of the arXiv dataset [7], which is accessible on Kaggle, for this investigation, paying particular attention to the abstract section of every data entry. After being retrieved, the abstracts were fed into a number of pre-trained models. Each model was required to produce an abstract summary, with the restriction that the summary may not be longer than two or three lines. This configuration guaranteed a uniform foundation for assessing how well various models performed in generating precise and logical summaries.

3.2. Generative pre-trained model (GPT) –

OpenAI's GPT (Generative Pre-trained Transformer) model is a state-of-the-art language model that makes use of the Transformer architecture. The GPT-3.5 version used in this study has a vast number of parameters (175 billion) and has been pre-trained on a wide range of web content. The model generates text by predicting the subsequent word in a sequence based on the context provided by the preceding words. The core of the GPT model is the Transformer architecture, which relies on self-attention mechanism.

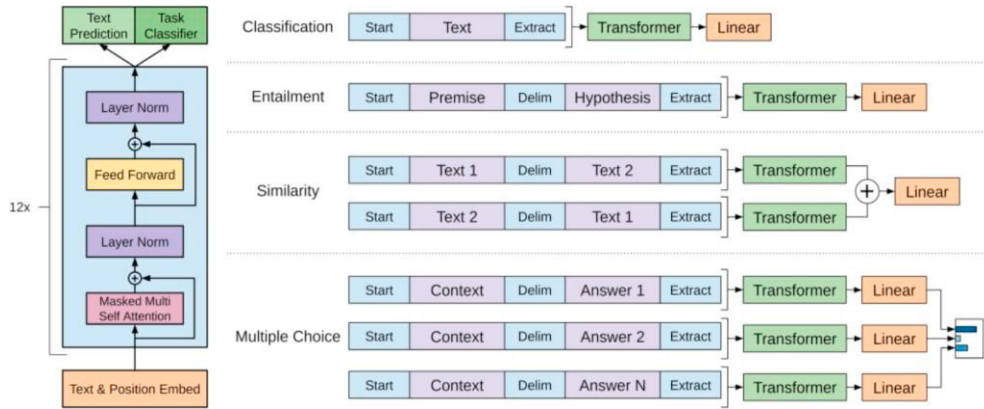


Fig 2. (left) Transformer architecture and training objectives used in this work. (right) Input transformations for fine-tuning on different tasks. We convert all structured inputs into token sequences to be processed by our pre-trained model, followed by a linear + softmax layer [28].

3.3. Large Language Model Meta AI (LLaMa) –

The LLaMa (Large Language Model Meta AI) model utilizes Transformer architecture concepts. LLaMa 3 enhances language model efficacy and scalability through improved architecture and training. LLaMa-13B, despite being $10\times$ smaller, surpasses GPT-3 in most benchmarks [29]. It incorporates a 2.5% sampling of arXiv LaTeX files for training [29]. Unlike GPT, which applies normalization to the architecture's output, LLaMa applies it to each transformer sub-layer's input [29].

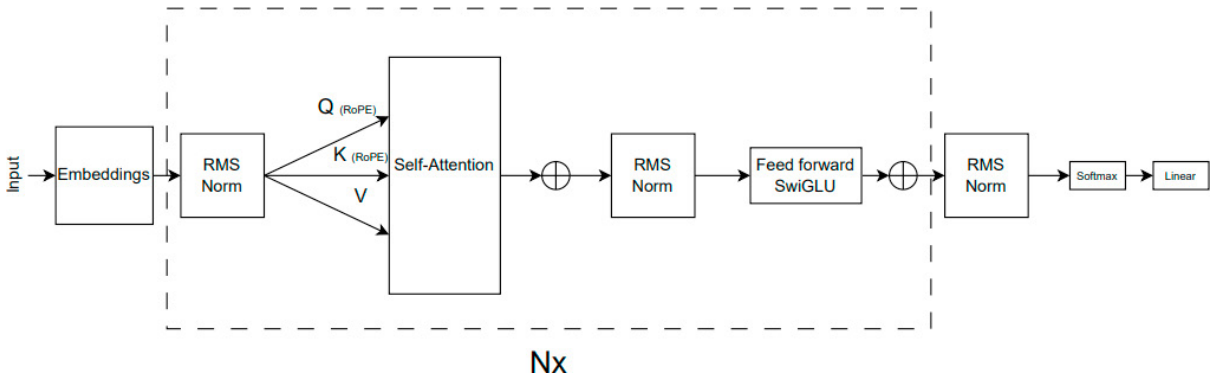


Fig 3. LLaMA inference Architecture.

LLaMA and other large language models are built on the Transformer architecture, comprising feed-forward neural networks and self-attention layers. The self-attention mechanism evaluates token relevance within the input sequence, enabling the model to capture long-range text dependencies. The core of the Transformer is the self-attention mechanism [29], which can be mathematically represented as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)[29]$$

Where:

- Q = query matrix.
- K = key matrix.
- V = value matrix.
- d_k = dimensionality of the keys.

LLaMa allows the model to focus on many input sequence segments simultaneously with the help of multi-head attention. This is done by splitting the query (Q), key (K), and value (V) matrices into multiple heads. After calculating the attention of each head independently, the results are concatenated [29].

$$\text{Multihead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W^O \quad (2)[29]$$

Where each head is calculated as,

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (3)[29]$$

3.4. Mixtral 8x7B

The Sparse Mixture of Experts (SMoE) language model Mixtral 8x7B introduces a new architecture based on Mistral 7B. It uses eight feedforward blocks (experts) per layer, with a router network dynamically selecting two experts per token for processing. This approach activates only 13 billion of the 47 billion accessible parameters during inference [30]. Trained with a 32,000-token context size, Mixtral surpasses Llama 2 70B and GPT-3.5 on various benchmarks, excelling in multilingualism, arithmetic, and code creation tasks.

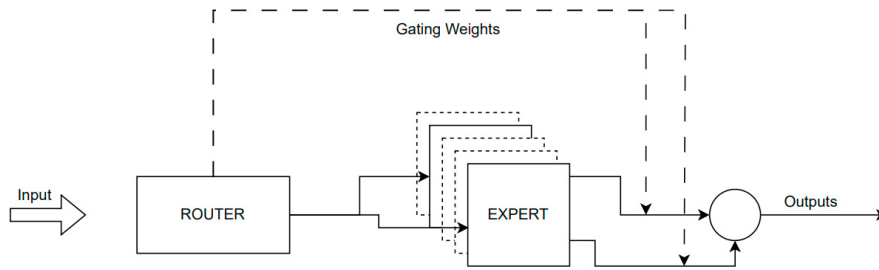


Fig 4. Mixture of Experts Layer.

The Mixtral 8x7B model's performance is enhanced by its "experts" layer, consisting of multiple feedforward blocks (experts) that handle input tokens as seen in figure 4. A router network dynamically selects two experts per token at each layer based on the context. These experts process the token and generate outputs. This design leverages 47 billion parameters while keeping only 13 billion active during inference, enabling efficient yet powerful language modeling capabilities.

3.5. Gemma

Gemma-2, the version implemented in this study, integrates an architecture that blends components of Transformer models and recurrent neural networks. The transformer architecture serves as the foundation for the actual architecture [31]. The basic parameters of the architecture are summarized in Table 2 [32].

Table 2: Key Model Parameters [32].

Parameters	2B	7B
d_model	2048	3072
Layers	18	28
Feedforward hidden dims	32768	49152
Num heads	8	16
Num KV heads	1	16
Head size	256	256
Vocab size	256128	256128

The self-attention mechanism in the Gemma model is augmented with a gating mechanism to control the flow of information [32], represented as:

$$\text{GatedAttention}(Q, K, V) = \sigma(G) \cdot \text{Attention}(Q, K, V) \quad (4)[32]$$

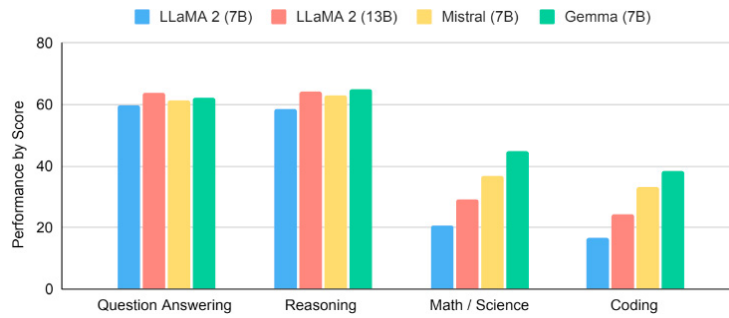


Fig 5. Comparison of Gemma with other models [32].

Figure 4 displays the results of comparing Gemma 7B's language understanding and generation performance across a range of capabilities to open models of similar size. After being categorized according to capability, the normal academic benchmark ratings are averaged.

3.6. Metrics

In order to statistically evaluate the performance of the models (GPT-3.5, LLaMa3, Mixtral 8x7B, and Gemma-2), a comparison of metrics - ROUGE-1, ROUGE-2, ROUGE-L, BLEU and BERT scores has been performed as shown in Table 3.

Table 3: Metric Description

Metric	Description	Good score
ROUGE – 1	Measures the unigram overlap for keyword inclusion.	0.5
ROUGE – 2	Measure the bigram overlap for sentence structure and explanation.	0.4
ROUGE – L	Measures the longest common subsequence overlap.	0.4
BLEU	Measures summary quality via precision comparison with references.	0.3 +
BERT	Evaluates semantic similarity between generated and reference summaries using BERT embeddings.	0.2

4. Results and Discussion

4.1. GPT 3.5

GPT-3.5 demonstrates good syntactical match shown in Table 4 with a median ROUGE-1 of 0.522, ROUGE-2 of 0.257, and ROUGE-L of 0.394, indicating it captures keywords, key phrases, and word sequences well. However, the maximum scores suggest potential over-lapping of n-grams. Its median BLEU score of 0.325 reflects solid performance. The model also achieves a median BERT score of 0.215, showing it captures semantics effectively, with scores between 0.157 and 0.289 confirming its strength in summarizing research papers.

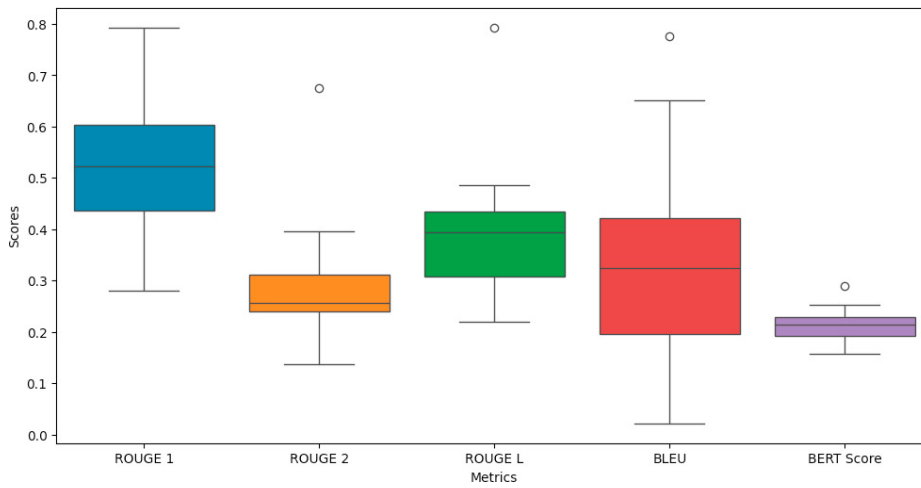


Fig 6. ROUGE, BLEU and BERT Scores of GPT

Table 4: GPT-3.5 test stats.

	ROUGE 1	ROUGE 2	ROUGE L	BLEU	BERT
Min	0.280303	0.137405	0.219697	0.021740	0.156724
Max	0.793220	0.675768	0.793220	0.776359	0.289074
q1	0.435986	0.240112	0.308605	0.194823	0.192153
median	0.522160	0.256729	0.393979	0.324752	0.214634
q3	0.602747	0.311986	0.434592	0.421636	0.227954

4.2 LLaMA 3

LLaMa 3.5 shown in Table 5 excellent syntactical match with median ROUGE-1 of 0.533, ROUGE-2 of 0.314, ROUGE-L of 0.403, and BLEU score of 0.322. Its scores range from 0.429 to 0.658 for ROUGE-1, 0.158 to 0.496 for ROUGE- 2, 0.244 to 0.559 for ROUGE-L, and 0.157 to 0.547 for BLEU, indicating that the model generates abstractive summaries, preserves semantics, and accurately captures key information without directly copying the reference summaries. Median BERT score of 0.197961 shows excellent semantic matches as well.

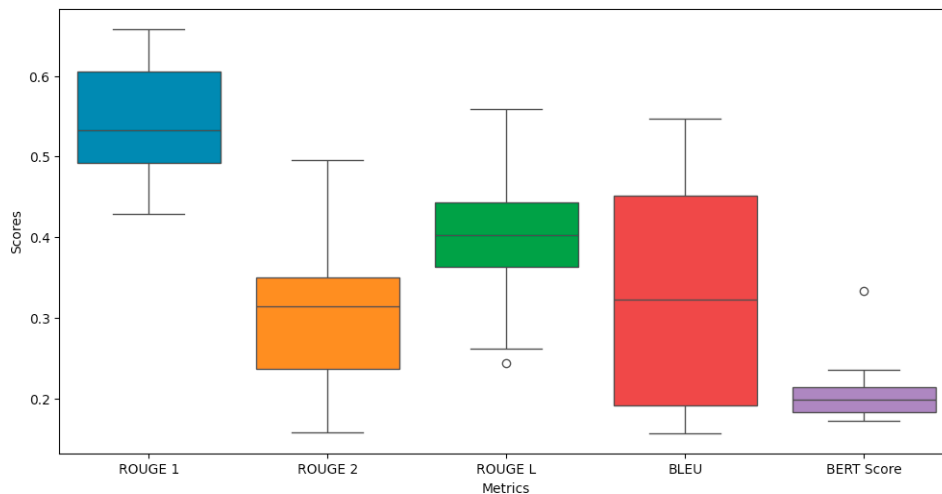


Fig 7. ROUGE, BLEU and BERT Scores of LLaMA

Table 5: LLaMA 3 test stats.

	ROUGE 1	ROUGE 2	ROUGE L	BLEU	BERT
min	0.428571	0.158273	0.244094	0.156912	0.172066
max	0.658333	0.495798	0.559242	0.546851	0.332933
q1	0.491777	0.236542	0.363331	0.190761	0.183126
median	0.532964	0.314300	0.402749	0.322132	0.197961
q3	0.605773	0.349901	0.442533	0.451107	0.213877

4.3 Mixtral 8x7B

According to the results shown in Table 6, Mixtral 8x7B continuously scores higher on each of these measures, proving its ability to reliably capture relevant content and efficiently summarise text from a purely quantitative perspective. But the ROUGE-1 score is pretty high, which can be inferred as the summary having too many words from the reference. Higher ROUGE – 2 indicates too many overlaps in phrases meaning that the summary generated might not be abstractive in nature. So, while the BERT scores are good indicating good semantic matches, the summaries are more extractive than abstractive.

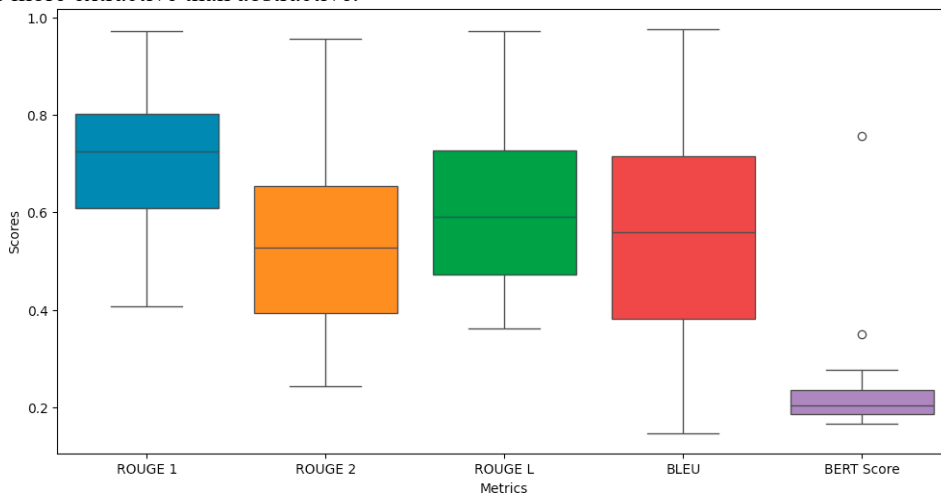


Fig 8. ROUGE, BLEU and BERT Scores of Mixtral

Table 6: Mixtral 8x7B test stats.

	ROUGE 1	ROUGE 2	ROUGE L	BLEU	BERT
min	0.407407	0.242991	0.361111	0.147306	0.167391
max	0.971722	0.956072	0.971722	0.976026	0.756747
q1	0.609260	0.393583	0.472741	0.380709	0.185781
median	0.725498	0.527328	0.591499	0.558624	0.204668
q3	0.802069	0.653728	0.727001	0.714389	0.234921

4.4 Gemma 2

Interestingly, Gemma-2 as in Table 7 had the lowest results out of the four models. The median scores for ROUGE and BLEU are lesser in comparison to LLaMa 3 and GPT 3.5. Even though it doesn't guarantee a bad overall score in terms of overlaps and syntax, it's definitely not the best. Lower BERT score also indicates lesser semantic matches which can lead to more nuanced summaries with more information loss, often resulting in ambiguous and incomplete summaries.

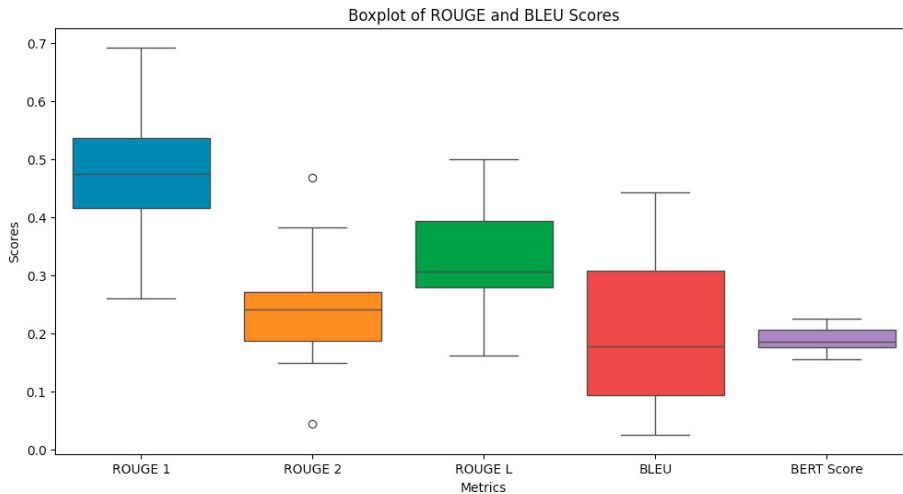


Fig 9. ROUGE, BLEU and BERT Scores of Gemma-2

Table 7: Gemma-2 test stats.

	ROUGE 1	ROUGE 2	ROUGE L	BLEU	BERT
min	0.259459	0.043716	0.162162	0.025553	0.154674
max	0.691765	0.468085	0.498824	0.441673	0.225367
q1	0.414801	0.187340	0.279025	0.094099	0.175324
median	0.474325	0.241046	0.306243	0.176832	0.186022
q3	0.536338	0.270742	0.393478	0.307129	0.205697

Overall, LLaMa 3 and GPT-3.5 generated summaries were more accurate in terms of context, more cohesive, and more representative of the main ideas of the original research paper with accurate syntax and meaning. The two models are good for avoiding nuances and retaining the information within the summary. As for Mixtral 8x7B, it might be more well suited for extractive summarisation tasks. Gemma-2 seems to be performing moderately on abstract summarisation task, but not as good as LLaMa 3 and GPT 3.5

5. Conclusion and Future Scope

While quantitative measures are essential, qualitative evaluations reveal that GPT-3.5 and LLaMa 3 excel at generating contextually accurate summaries. This highlights the importance of combining automated metrics with human assessments for a comprehensive understanding of a model's summarization abilities. Gemma-2's lower results emphasize the limitations of relying solely on numerical assessments, despite its ability to produce logical summaries. Future research should aim for broader frameworks that combine quantitative and qualitative evaluations to better reflect LLMs' true capabilities in handling complex summarization tasks.

Meta's LLaMA model excels in both performance and generating clear, well-structured summaries, making it a strong tool for summarization tasks. Future research will explore additional models for specialized domains like scientific, medical, and legal texts, aiming to assess their broader applicability and identify areas for optimization.

References

[1] Van Veen, D., Van Uden, C., Blankemeier, L., Delbrouck, J. B., Aali, A., Bluethgen, C., ... & Chaudhari, A. S. (2024). Adapted large language models can outperform medical experts in clinical text summarization. *Nature medicine*, 30(4), 1134-1142.

- [2] Langston, O., & Ashford, B. (2024). Automated summarization of multiple document abstracts and contents using large language models. *Authorea Preprints*.
- [3] Sengar, S. S., Hasan, A. B., Kumar, S., & Carroll, F. (2024). Generative artificial intelligence: a systematic review and applications. *Multimedia Tools and Applications*, 1-40.
- [4] Papineni, K., Roukos, S., Ward, T., & Zhu, W. J. (2002, July). Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics* (pp. 311-318).
- [5] Lin, C. Y. (2004, July). Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out* (pp. 74-81).
- [6] Kenton, J. D. M. W. C., & Toutanova, L. K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of naacL-HLT* (Vol. 1, p. 2).
- [7] Clement, C. B., Bierbaum, M., O'Keeffe, K. P., & Alemi, A. A. (2019). On the use of arxiv as a dataset. *arXiv preprint arXiv:1905.00075*.
- [8] Keswani, G., Bisen, W., Padwad, H., Wankhedkar, Y., Pandey, S., & Soni, A. (2024). Abstractive Long Text Summarization Using Large Language Models. *Int. J. Intell. Syst. Appl. Eng*, 12, 160-168.
- [9] Yadav, A.K., Ranvijay, Yadav, R.S. and Maurya, A.K., 2024. Graph-based extractive text summarization based on single document. *Multimedia Tools and Applications*, 83(7), pp.18987-19013.
- [10] Aswani, S., Choudhary, K., Shetty, S., & Nur, N. (2024). Automatic text summarization of scientific articles using transformers—A brief review. *Journal of Autonomous Intelligence*, 7(5).
- [11] Jin, H., Zhang, Y., Meng, D., Wang, J., & Tan, J. (2024). A comprehensive survey on process-oriented automatic text summarization with exploration of llm-based methods. *arXiv preprint arXiv:2403.02901*.
- [12] Alli-Balogun, M. Application of Text Summarization on Text-Based Generative Adversarial Networks. *International Journal of Computer (IJC)*, 50(1), 8-31.
- [13] Han, C., Feng, J., & Qi, H. (2024). Topic model for long document extractive summarization with sentence-level features and dynamic memory unit. *Expert Systems with Applications*, 238, 121873.
- [14] Liu, W., Sun, Y., Yu, B., Wang, H., Peng, Q., Hou, M., ... & Liu, C. (2024). Automatic Text Summarization Method Based on Improved TextRank Algorithm and K-Means Clustering. *Knowledge-Based Systems*, 287, 111447.
- [15] Sun, Y., & Platoš, J. (2024). Abstractive text summarization model combining a hierarchical attention mechanism and multiobjective reinforcement learning. *Expert Systems with Applications*, 248, 123356.
- [16] Yadav, N., & Gopinathan, D. (2024). Graph Based Analytics and Review of Knowledge Graph Field by Using Two Decade Data-Finding Paradigms and Opportunities.
- [17] Surjandari, I., Gunawijaya, H. P., & Puspita, A. N. G. (2024). Topic-Based Segmentation in Email Marketing.
- [18] Yadav, P., Kashyap, I., & Bhati, B. S. (2024). Impact of Double Negation through Majority Voting of Machine Learning Algorithms.
- [19] Glickman, M., & Zhang, Y. (2024). AI and generative AI for research discovery and summarization. *Harvard Data Science Review*, 6(2).
- [20] Pesaru, A., Gill, T. S., & Tangella, A. R. (2023). AI assistant for document management Using Lang Chain and Pinecone. *International Research Journal of Modernization in Engineering Technology and Science*, 5(6), 3980-3983.
- [21] Nova, K. (2023). Generative AI in healthcare: advancements in electronic health records, facilitating medical languages, and personalized patient care. *Journal of Advanced Analytics in Healthcare Management*, 7(1), 115-131.
- [22] Bandi, A., Adapa, P. V. S. R., & Kuchi, Y. E. V. P. K. (2023). The power of generative ai: A review of requirements, models, input-output formats, evaluation metrics, and challenges. *Future Internet*, 15(8), 260.
- [23] Huerta-Soto, R., Francis, F., Asis-López, M., & Panduro-Ramirez, J. (2023). Implementation of machine learning in Supply Chain Management Process for sustainable development by multiple regression analysis approach (MRAA).
- [24] Kim, S., & Yoon, B. (2022). Multi-document summarization for patent documents based on generative adversarial network. *Expert Systems with Applications*, 207, 117983.
- [25] Altmami, N. I., & Menai, M. E. B. (2022). Automatic summarization of scientific articles: A survey. *Journal of King Saud University-Computer and Information Sciences*, 34(4), 1011-1028.
- [26] Tarannum, S., Sonar, P., & Khaimar, A. A. K. (2021). NLP based Text Summarization Techniques for News Articles: Approaches and Challenges. *International Research Journal of Engineering and Technology (IRJET)* Volume, 8.
- [27] Gao, Y., Meyer, C. M., & Gurevych, I. (2020). Preference-based interactive multi-document summarisation. *Information Retrieval Journal*, 23(6), 555-585.
- [28] Radford, A. (2018). Improving language understanding by generative pre-training.
- [29] Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M. A., Lacroix, T., ... & Lample, G. (2023). Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.
- [30] Jiang, A. Q., Sablayrolles, A., Roux, A., Mensch, A., Savary, B., Bamford, C., ... & Sayed, W. E. (2024). Mixtral of experts. *arXiv preprint arXiv:2401.04088*.
- [31] Vaswani, A. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*.
- [32] Team, G., Mesnard, T., Hardin, C., Dadashi, R., Bhupatiraju, S., Pathak, S., ... & Kenealy, K. (2024). Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*.